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SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 3

Comparative Analysis

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA0403**
SLOT: A

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

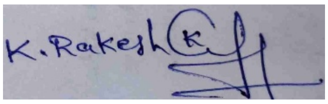
CO Weightage: 25% **Duration: 90 minutes**

Assessment Tool Weightage: 40% **Marks: 25**

Code of Conduct:

I Kothakota Rakesh / 192525075 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student **Name of the student: K. Rakesh**

Criteria	Selection of Studies/Parameters (2)	Comparison & Differentiation (2.5)	Critical Evaluation (2.5)	Synthesis & Insight Generation (2)	Clarity & Presentation *(1)*	Total (10)
Weightage	*20 %*	*25%*	*25%*	*20 %*	*10 %*	**100 %**
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Signature of the Faculty: ***Total Marks:***

QUESTIONS: 05 **Total Marks:25**

Q1. CNN vs Fully Connected Neural Network

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

CNN vs Fully Connected Neural Network

For classifying X-ray images, a Convolutional Neural Network (CNN) and a traditional Fully Connected Neural Network (FCNN) differ fundamentally in how they process image data:

Aspect	Fully Connected Neural Network	Convolutional Neural Network
Architectural Design	Every neuron in one layer connects to every neuron in the next; image is flattened into a 1D vector before input.	Uses local receptive fields with convolutional layers, filters, and pooling layers; preserves the 2D spatial structure of the image.
Motivation	General-purpose approximator, not designed with spatial data in mind.	Specifically designed to exploit spatial locality and hierarchical structure present in images.
Spatial Feature Extraction	Flattening destroys spatial relationships between neighboring pixels.	Convolutional filters scan local neighborhoods, preserving and extracting spatial patterns such as edges, textures, and shapes.
Number of Parameters	Extremely high — a 224x224x3 image connected to a single dense layer of 100 neurons needs over 15 million weights.	Much lower — filters (e.g., 3x3xC) are shared across the entire image, independent of image size.
Computational Complexity	High computation and memory cost, scales poorly with image resolution.	More computationally efficient due to weight sharing and sparse local connections.
Ability to Handle Image Data	Poor — loses spatial context, prone to overfitting, cannot generalize well to shifted or scaled features.	Excellent — naturally suited to grid-like image data, robust to minor translations and distortions.

Role of Convolutional Layers, Filters, and Pooling Layers

- Convolutional Layers: Slide learnable filters over the image to detect local patterns (edges, textures, abnormal regions in an X-ray) while preserving spatial relationships between pixels.
- Filters: Small weight matrices that specialize in detecting specific visual cues; multiple filters per layer allow the network to capture a rich variety of low- to high-level features.
- Pooling Layers: Downsample feature maps (e.g., via max pooling), reducing dimensionality, computational cost, and providing translation invariance so that a feature is still recognized even if it shifts slightly within the image.

Why CNNs Are More Suitable for Image Classification

CNNs are better suited for image classification tasks such as X-ray analysis because they preserve spatial structure, drastically reduce the number of trainable parameters through weight sharing, extract hierarchical features automatically without manual feature engineering, and generalize better to variations in position, scale, and orientation of key visual patterns. This makes CNNs far more accurate, efficient, and scalable than fully connected networks for image-based diagnostic tasks.

Q2. Different CNN Layers and Filters

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of **convolutional layers, pooling layers, activation layers, and fully connected layers** in terms of feature extraction, spatial information, computational requirements, and classification. Further compare **early-layer filters and deeper-layer filters** based on the type of features they learn, and justify how these layers collectively improve defect detection.

Different CNN Layers and Filters

In a CNN-based defect detection system for manufacturing, each layer type plays a distinct role:

Layer Type	**Feature Extraction Role**	**Spatial Information**	**Computational Requirement**	**Role in Classification**
Convolutional Layer	Extracts local features (edges, textures, scratches, cracks) using learnable filters.	Preserves spatial structure of the product image.	Moderate — depends on filter size, number of filters, and stride.	Provides the feature maps used by deeper layers to identify defects.
Pooling Layer	Aggregates and summarizes nearby feature activations (e.g., max pooling).	Reduces spatial resolution while retaining dominant features.	Low — no learnable parameters, only downsampling.	Reduces overfitting and computational load before classification.
Activation Layer (ReLU)	Introduces non-linearity, allowing detection of complex defect patterns.	Does not alter spatial dimensions.	Very low — simple element-wise operation.	Enables the network to learn non-linear decision boundaries between defective and non-defective products.
Fully Connected Layer	Combines high-level features into a global representation.	Spatial information is flattened/lost at this stage.	High — dense connections between all neurons.	Produces the final defect/no-defect classification decision.

Early-Layer Filters vs Deeper-Layer Filters

Aspect	Early-Layer Filters	Deeper-Layer Filters
Type of Features Learned	Low-level features: edges, corners, color gradients, simple textures.	High-level, abstract features: complex defect shapes, specific patterns unique to defect categories.
Receptive Field	Small, localized region of the image.	Large, covering broader context across the image.
Generalizability	Features tend to be generic and reusable across different tasks.	Features are more task-specific, tuned to the exact defect types being detected.

Collectively, early layers detect basic visual irregularities (edges of a scratch or dent), middle layers combine these into meaningful shapes (crack patterns, surface discoloration regions), and deeper layers assemble them into complete, high-level defect representations. Pooling and activation layers make this hierarchical extraction efficient and non-linear, while the fully connected layer integrates all the learned evidence to output an accurate defect classification, improving detection reliability and reducing false positives/negatives in the manufacturing pipeline.

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare **parameter sharing in CNNs with independent weights in fully connected networks** in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for large-scale image-processing applications.

For a smart surveillance system processing thousands of high-resolution images, the choice between CNN-style parameter sharing and the independent weights of a fully connected network has major practical implications:

Aspect	Independent Weights (Fully Connected)	Parameter Sharing (CNN)
Number of Parameters	Very high; grows proportionally with image resolution and number of neurons.	Low and independent of image size; determined only by filter size and number of filters.
Memory Requirements	Large memory footprint, difficult to scale to high-resolution surveillance feeds.	Compact memory footprint, well suited for processing large volumes of high-resolution images.
Computational Complexity	High — every input pixel multiplies with a unique weight for every neuron.	Lower — the same small filter is reused (convolved) across all spatial locations.
Feature Detection	Learns a unique, position-specific pattern for every input location; cannot generalize a pattern learned in one region to another.	Learns a single reusable pattern detector (e.g., an edge or object-part detector) applicable everywhere in the image.

Translation-Related Invariance	None — a shifted object is treated as an entirely new pattern.	Strong — a filter detects the same feature (e.g., a person's silhouette) regardless of where it appears in the frame.

Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering **AlexNet and ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

How Weight Reuse Improves Learning Efficiency

Because a single filter's weights are reused across every spatial location in the image, the network needs to learn far fewer unique parameters. This makes training faster, reduces the risk of overfitting (particularly valuable when labelled surveillance data is limited), and lowers the memory and compute needed for both training and real-time inference. Additionally, since the filter is applied uniformly, a person or object detected at one position in the frame is recognized just as reliably if it appears elsewhere — a property essential for surveillance systems where subjects move freely across the camera's field of view.

Importance for Large-Scale Image Processing

For large-scale, real-time surveillance applications, parameter sharing is essential: it enables the system to process high volumes of high-resolution frames within strict latency budgets, run efficiently on edge devices with limited memory/compute, and maintain robust detection performance regardless of where a person or object appears in the frame — making CNNs the practical and scalable choice over independent-weight fully connected networks.

Criterion	**AlexNet**	**ResNet**
Network Depth	Shallow — 8 layers (5 convolutional + 3 fully connected).	Very deep — commonly 18, 34, 50, 101, or 152 layers.
Layer Organization	Simple sequential stack of conv, pooling, and FC layers.	Organized into residual blocks with identity/skip connections between layers.
Convolutional Filters	Uses large filters (e.g., 11x11, 5x5) in early layers.	Uses smaller filters (mostly 3x3, with 1x1 bottleneck filters), enabling deeper stacking at lower cost.
Parameter Requirements	~60 million parameters, mostly concentrated in large fully connected layers.	Fewer parameters relative to depth due to bottleneck design and global average pooling instead of large FC layers.
Feature Extraction Capability	Limited hierarchical depth restricts the complexity of features learned.	Very strong — deep hierarchy captures rich low-level to highly abstract features.
Training Difficulty	Easier to train due to shallow depth,	Deeper networks are inherently harder

	but capacity-limited.	to optimize, but residual connections make training practical and stable.
Vanishing-Gradient Problem	Becomes significant if depth is increased beyond the original design.	Effectively mitigated — skip connections allow gradients to flow directly through the network.
Computational Cost	Lower per-layer cost but large FC layers add significant compute/memory.	Higher due to depth, but efficient bottleneck blocks keep the added cost proportionally manageable.
Classification Performance	Achieved strong results in 2012 (ImageNet) but is outperformed by deeper modern architectures.	Consistently achieves higher accuracy on complex, large-scale image classification benchmarks.

Recommendation

For a complex image recognition application, ResNet is the recommended architecture. Its residual connections solve the vanishing-gradient problem that limits very deep networks, allowing it to be trained much deeper than AlexNet and thereby learn significantly richer and more discriminative features. Although ResNet has a higher raw computational cost due to its depth, its parameter-efficient bottleneck design and superior, well-documented classification accuracy on complex datasets make it the more suitable and scalable choice over AlexNet for demanding, real-world computer vision applications.

Q5. Regularized CNN vs Non-Regularized CNN

An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization and a CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

For the agricultural organization's plant disease classifier — which shows high training accuracy but poor performance on unseen leaf images (a classic sign of overfitting) — comparing a non-regularized CNN with a regularized CNN highlights the necessary fix:

Aspect	**CNN Without Regularization**	**CNN With Regularization (Dropout, Augmentation, BatchNorm)**
Overfitting	High risk — model memorizes training images, including noise and irrelevant background details.	Substantially reduced — regularization techniques actively discourage memorization of training-specific patterns.
Generalization	Poor performance on new, unseen leaf images from different conditions (lighting, angle, background).	Improved generalization to real-world, previously unseen field images.
Training Stability	Can be unstable with fluctuating validation accuracy/loss as the model	More stable training with smoother convergence, especially with Batch

	overfits.	Normalization.
Computational Requirements	Lower per-epoch cost, but often requires more retraining/tuning to fix overfitting after the fact.	Slightly higher per-epoch cost (extra computation for dropout masks, augmented samples, normalization), but more efficient overall due to fewer wasted training cycles.
Model Performance (Real-World)	High training accuracy but low, unreliable real-world/field accuracy.	More balanced training and validation accuracy, leading to reliable, trustworthy field performance.

Which Regularization Techniques Should Be Applied

- Data Augmentation: Apply rotations, flips, zoom, brightness/contrast variation, and cropping to simulate the diverse field conditions (lighting, leaf orientation, background) under which real images will be captured, artificially expanding a limited disease-image dataset.
- Dropout: Randomly deactivate neurons in fully connected layers during training to prevent the network from over-relying on specific feature combinations tied to the training set.
- Batch Normalization: Normalize layer activations to stabilize and speed up training, allow higher learning rates, and provide a mild additional regularization effect.
- Additional recommended measures: Early stopping based on validation loss and, where possible, transfer learning from a model pretrained on a larger plant/leaf image dataset.

Justification for the Most Appropriate Approach

Given that the model already shows a clear overfitting symptom (high training accuracy, poor unseen-image accuracy), the most appropriate approach is to combine all three regularization techniques rather than relying on a single one: data augmentation addresses the limited diversity of training images (a common issue in agricultural datasets), dropout prevents the fully connected layers from memorizing training-specific patterns, and batch normalization stabilizes training while offering additional regularization. Using these together produces a CNN that generalizes reliably to new, real-world leaf images captured under varying field conditions — which is essential for a disease-classification system intended for practical agricultural deployment.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Selection of Studies/Parameters	**20%**	**Selects highly relevant and diverse studies with well-defined comparison parameters**	**Selects relevant studies with minor gaps in parameter definition**	**Limited or partially relevant selection**	**Poor or inappropriate selection of studies**
Comparison & Differentiation	**25%**	**Clearly compares multiple aspects (methods, results, limitations) with strong differentiation**	**Good comparison with minor gaps**	**Basic comparison with limited depth**	**No clear comparison; mostly descriptive**
Critical Evaluation	**25%**	**Critically evaluates strengths, weaknesses, and implications with strong justification**	**Provides evaluation with some justification**	**Limited evaluation; lacks depth**	**No critical evaluation provided**
Synthesis & Insight Generation	**20%**	**Generates meaningful insights and conclusions from comparisons**	**Some insights derived from comparison**	**Limited or unclear insights**	**No insights generated**
Clarity	**10%**	**Information is clearly presented (tables/figures) with logical structure**	**Generally clear with minor issues**	**Presentation lacks clarity or structure**	**Poor presentation and difficult to understand**